

$$\therefore \begin{bmatrix} 3 & 5 & -4 \\ -3 & -2 & 4 \\ 6 & 1 & -8 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad (\text{By defn. of } Ax=b)$$

$\therefore$ The equation is of the form $Ax=0$.

To verify whether this equation has a nontrivial solution, we must check whether the equation has ^{at least} one free variable.

- By defn, a variable of a system of linear equations is free if its corresponding column in the ~~augmented~~ ^{coefficient} matrix has no pivot positions.
- By defn, a pivot position of the matrix corresponds to a leading 1 of the given row in reduced echelon form.

$$\begin{array}{l} \begin{bmatrix} 3 & 5 & -4 \\ -3 & -2 & 4 \\ 6 & 1 & -8 \end{bmatrix} \xrightarrow[\text{Row 3 = Row 3 + 3 \times Row 2}]{\text{Row 2 = Row 2} + \text{Row 1}} \begin{bmatrix} 3 & 5 & -4 \\ 0 & 3 & 0 \\ 6 & 1 & -8 \end{bmatrix} \xrightarrow[\text{-2 \times Row 1}]{\text{Row 3 = Row 3}} \begin{bmatrix} 3 & 5 & -4 \\ 0 & 3 & 0 \\ 0 & -9 & 0 \end{bmatrix} \\ \xrightarrow[\text{3 \times Row 2}]{\text{Row 3 = Row 3} + \text{3 \times Row 2}} \begin{bmatrix} 3 & 5 & -4 \\ 0 & 3 & 0 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow[\text{1/3}]{\text{Row 2 = Row 2} \times} \begin{bmatrix} 3 & 5 & -4 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow[\text{-5 \times Row 2}]{\text{Row 1 = Row 1}} \begin{bmatrix} 3 & 5 & -4 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \\ \xrightarrow[\text{\times 1/3}]{\text{Row 1 = Row 1}} \begin{bmatrix} 1 & 0 & -4/3 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \end{array}$$

Verification of whether the final matrix is in reduced row echelon form.

1. Row 1 and Row 2 are non-zero rows. Row 3 is a zero row. $\therefore$ The condition that all the non-zero rows must be above the zero rows is True.
2. Row 1 Leading Entry = 1 in Column 1
Row 2 Leading Entry = 1 in Column 2

$\therefore$ The condition that each row must have its leading entry in a column to the right of the columns of the leading entries of all rows above it is True.